

DUNCAN CAMERON-STEINKE

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AI researcher and recent MSc graduate in Mathematics and Statistics, specializing in uncertainty-aware deep learning and volatility forecasting with hidden Markov models.

Three years of experience in data and software engineering, most recently as a senior data engineer at ServiceTitan, plus two years as co-founder and CTO of an analytics startup.

EXPERIENCE

ASSOCIATE RESEARCHER

Sep 2026 – Present 2 mos

The University of British Columbia · Kelowna, BC

- Conducting full-time research in statistics and artificial intelligence, studying uncertainty quantification in deep learning.
- Developing Belief Distribution Learning, a novel deep learning framework with learned predictive uncertainty.
- Co-authoring a research paper on this framework with Dr. John Thompson, including empirical validation on benchmark datasets.

SENIOR DATA ENGINEER

Aug 2022 – Apr 2024 1 yr 9 mos

ServiceTitan

- Embedded with executive, business development, and customer success teams to turn open business problems into shipped analytics.
- Architected the data model and owned the pipelines behind a client-facing analytics product (Snowflake, SQL) serving 1,800+ customers.
- Designed the A/B testing methodology for an AI chat assistant automating call center support, using sample sizing and significance testing to measure impact on agent handling time.

ANALYTICS ENGINEER

Jul 2021 – Aug 2022 1 yr 2 mos

HomeX / Schedule Engine (acquired by ServiceTitan)

- Established a single source of truth for reporting, defining business metrics where none existed and building the data analytics platform.

CO-FOUNDER & CTO

Jun 2019 – Jul 2021 2 yrs 2 mos

Backr.ai (YouTube analytics startup)

- Co-founded a YouTube analytics platform and grew it to 50 customers, including channels with 3M+ subscribers.
- Developed a custom NLP model to classify YouTube comments and analyze sentiment, helping creators grow engagement on their channels.
- Hired and led a team of three full-time developers and a data analyst, owning product roadmap and technical strategy from inception.

🏆 Won first prize (\$30K) at the DDQIC pitch competition · 🏆 NextAI Montreal AI accelerator (HEC Montréal, Mila)

MACHINE LEARNING RESEARCH INTERN

Jan 2019 – Apr 2019 4 mos

Institute for Infocomm Research · Singapore

- Authored an NLP research paper (arXiv preprint) on unsupervised text expansion from pre-trained embeddings.
- Improved short-text classification accuracy under limited training data via pseudo-documents.

Research paper (arXiv) (<https://arxiv.org/abs/1909.07512>) · 🏆 Singapore International Pre-Graduate Award

EDUCATION

THE UNIVERSITY OF BRITISH COLUMBIA

Sep 2024 – Aug 2026

Master of Science (MSc) · Kelowna, BC

Mathematics and Statistics

GPA: 4.33/4.33

- Coursework in nonparametric models, stochastic processes, differential equations, generalized linear models, and optimization.

QUEEN'S UNIVERSITY

Sep 2015 – May 2020

Bachelor of Applied Science (BASc) · Kingston, ON

Engineering Physics

GPA: 3.9/4.0

- Core studies in physics, mathematics, statistics, and computer programming.

🏆 Dean's Scholar · 🏆 H. Jansen Memorial Scholarship (highest standing in Physics I & II)

CERTIFICATIONS

MIT SCHWARZMAN COLLEGE OF COMPUTING

2022

Professional Certificate · Remote

Data Science and Machine Learning: Making Data-Driven Decisions

RESEARCH PROJECTS

BELIEF DISTRIBUTION LEARNING

Mar 2026 – Present

Research paper (in preparation)

- Created a novel deep learning algorithm grounded in belief theory by combining Subjective Logic with Evidential Deep Learning.
- Introduced per-label uncertainty modeling using Dirichlet distributions.
- Evaluated against published Label Distribution Learning algorithms, demonstrating improved accuracy and better-calibrated uncertainty estimates.

Python TensorFlow Keras Machine Learning Deep Learning Uncertainty Quantification

VOLATILITY FORECASTING WITH UNCERTAINTY-AWARE COVARIATES

Jul 2025 – Apr 2026

Research project

- Added forecasting to the hidden Markov model package hmmTMB.
- Coupled an HMM with an uncertainty-aware deep learning model using a custom loss function.
- Demonstrated improved forecasting of volatility regimes in the TSX composite index compared to HMM or deep learning alone.

Python R Hidden Markov Models Time Series Forecasting Deep Learning

FORECASTING RADIO RECOMBINATION LINE SCIENCE

Sep 2025 – Jul 2026

MSc thesis, UBC Okanagan

- Scientific research project predicting the performance of a next-generation radio telescope.
- Extrapolated science results given the telescope design and existing surveys as priors.
- Thesis is available at <https://duncancs.ca>.

Python Statistical Modeling Statistics Radio Astronomy

SKILLS

A/B Testing · AWS · Business Intelligence · Data Engineering · Data Pipelines · Deep Learning · Entrepreneurship · Full Stack · GCP · Hidden Markov Models · Keras · Leadership · Machine Learning · NLP · Product Management · Python · R · Radio Astronomy · Snowflake · SQL · Statistical Modeling · Statistics · Tableau · TensorFlow · Time Series Forecasting · Uncertainty Quantification